NAT: Network Address Translation

Private Network

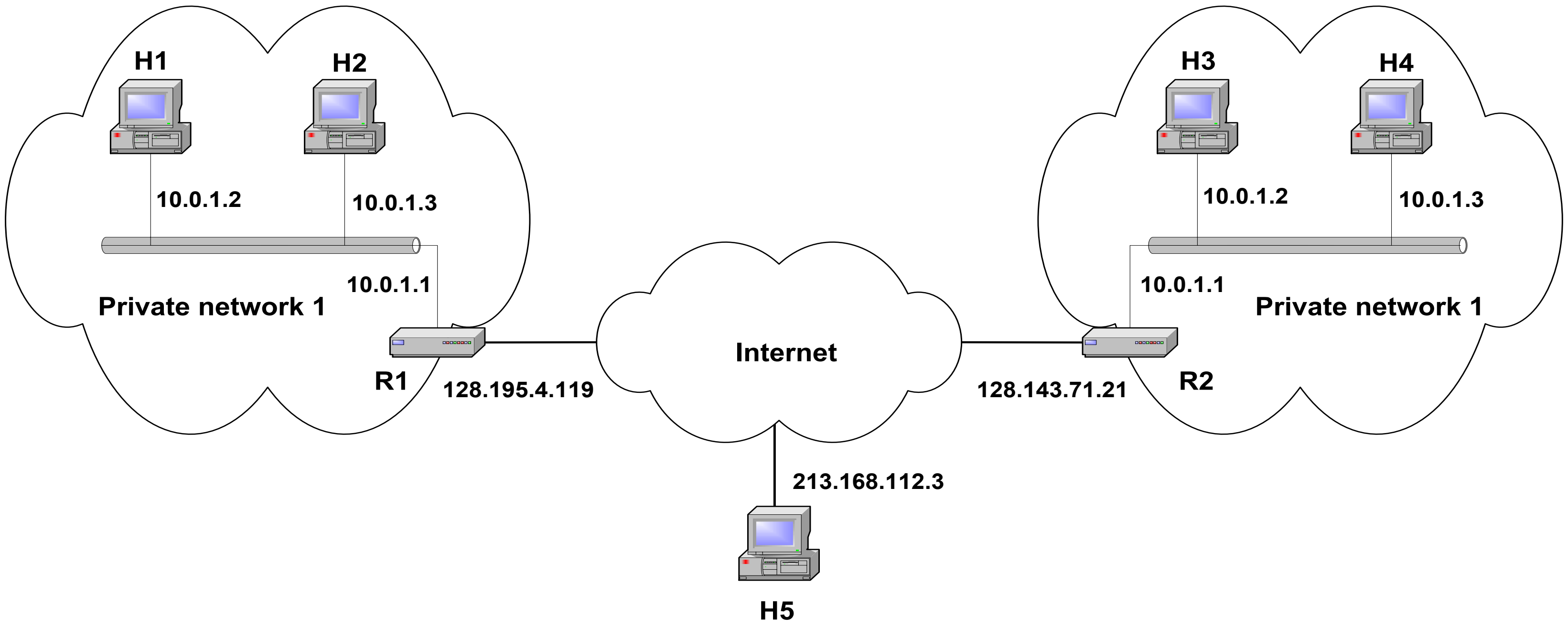
- *Private IP* network is an IP network that is not directly connected to the Internet

- IP addresses in a private network can be assigned arbitrarily.
  - Not registered and not guaranteed to be globally unique

- Generally, private networks use addresses from the following experimental address ranges (**Free and *non-routable addresses***): **[RFC 1918]**
  - 10.0.0.0 – 10.255.255.255
  - 172.16.0.0 – 172.31.255.255
  - 192.168.0.0 – 192.168.255.255

Class	RFC 1918 Internal Address Range	CIDR Prefix
A	10.0.0.0 - 10.255.255.255	10.0.0.0/8
B	172.16.0.0 - 172.31.255.255	172.16.0.0/12
C	192.168.0.0 - 192.168.255.255	192.168.0.0/16

Private Addresses



Network Address Translation (NAT)

- RFC 1631

- A short term solution to the problem of the depletion of IP addresses
  - Long term solution is IP v6
  - CIDR (Classless Inter Domain Routing) is a possible short term solution
  - NAT is another

- NAT is a way to conserve IP addresses
  - Can be used to hide a number of hosts behind a single IP address
  - Uses private addresses:
    - 10.0.0.0-10.255.255.255,
    - 172.16.0.0-172.32.255.255 or
    - 192.168.0.0-192.168.255.255

Network Address Translation (NAT)

- NAT is a router function where IP addresses (and possibly port numbers) of IP datagrams are replaced at the boundary of a private network

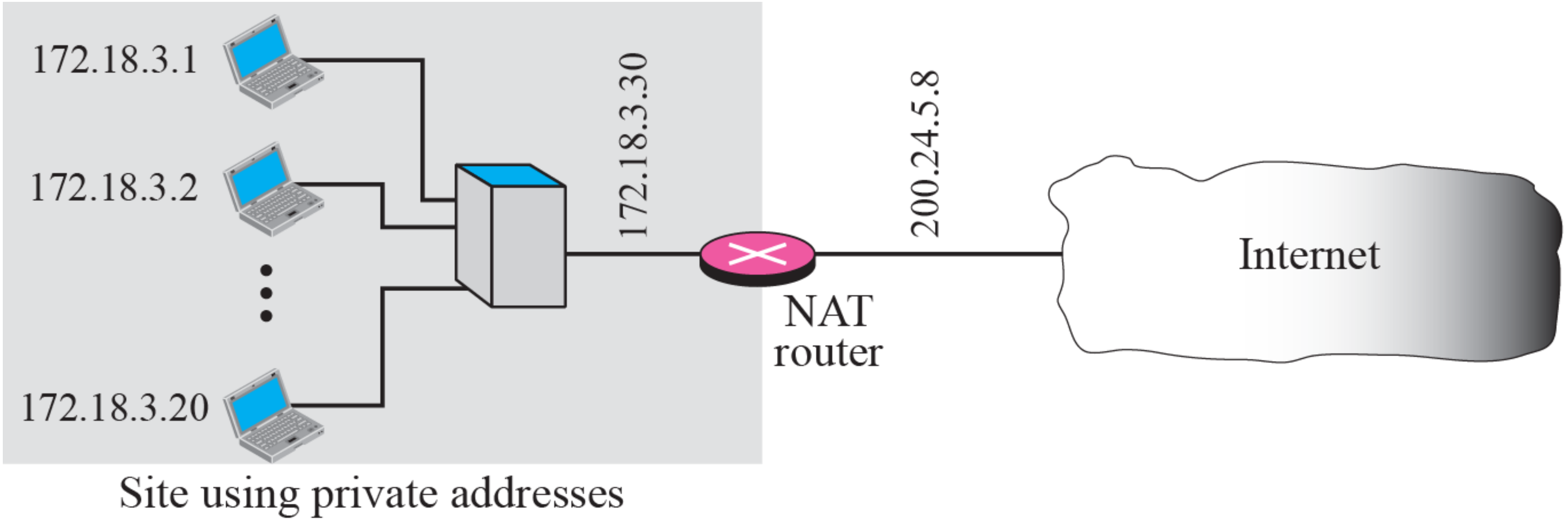
- NAT is a method that enables hosts on private networks to communicate with hosts on the Internet

- NAT is run on routers that connect private networks to the public Internet, to replace the IP address-port pair of an IP packet with another IP address-port pair.

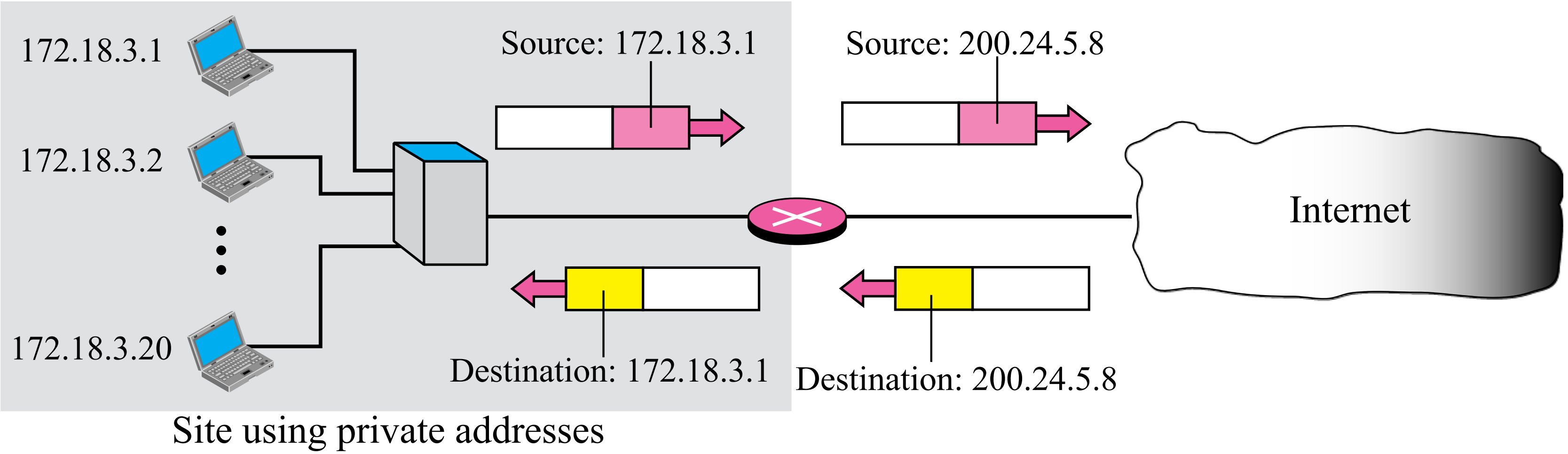
- NAT translates non-routable, private, internal addresses into routable, public addresses.

- NAT has an added benefit of adding a degree of privacy and security to a network because it hides internal IP addresses from outside networks.

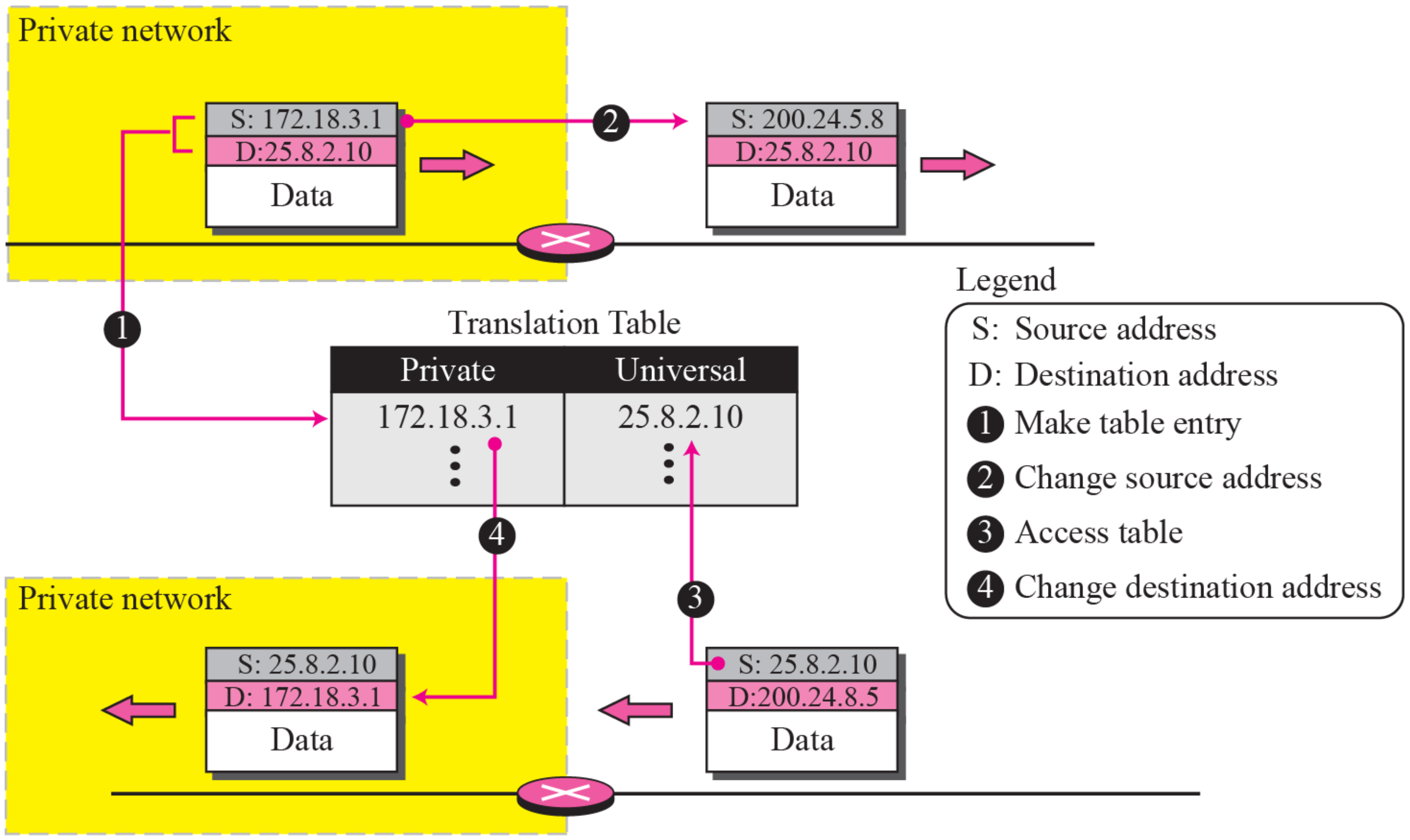
Network Address Translation (NAT)



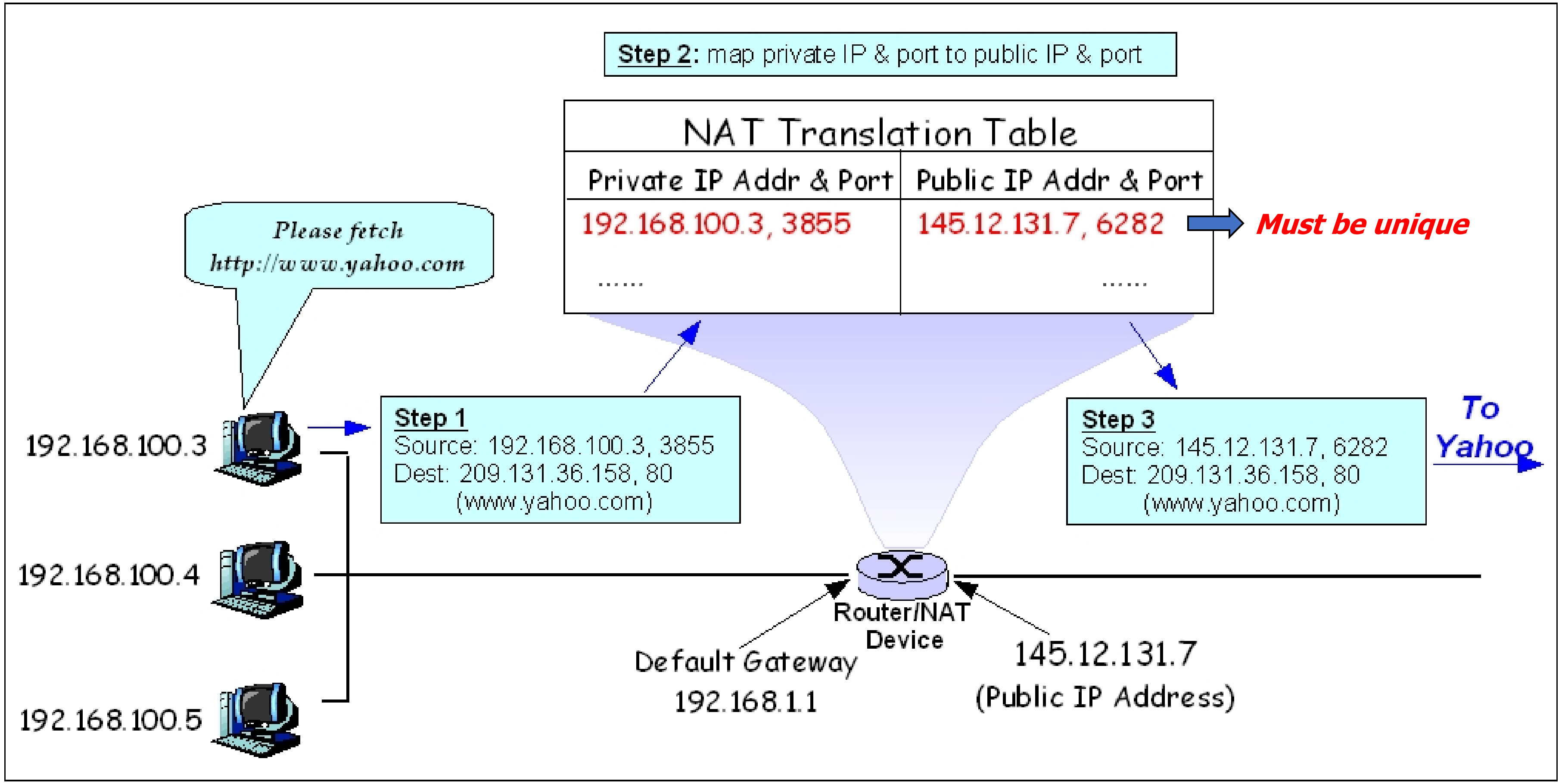
Address Resolution



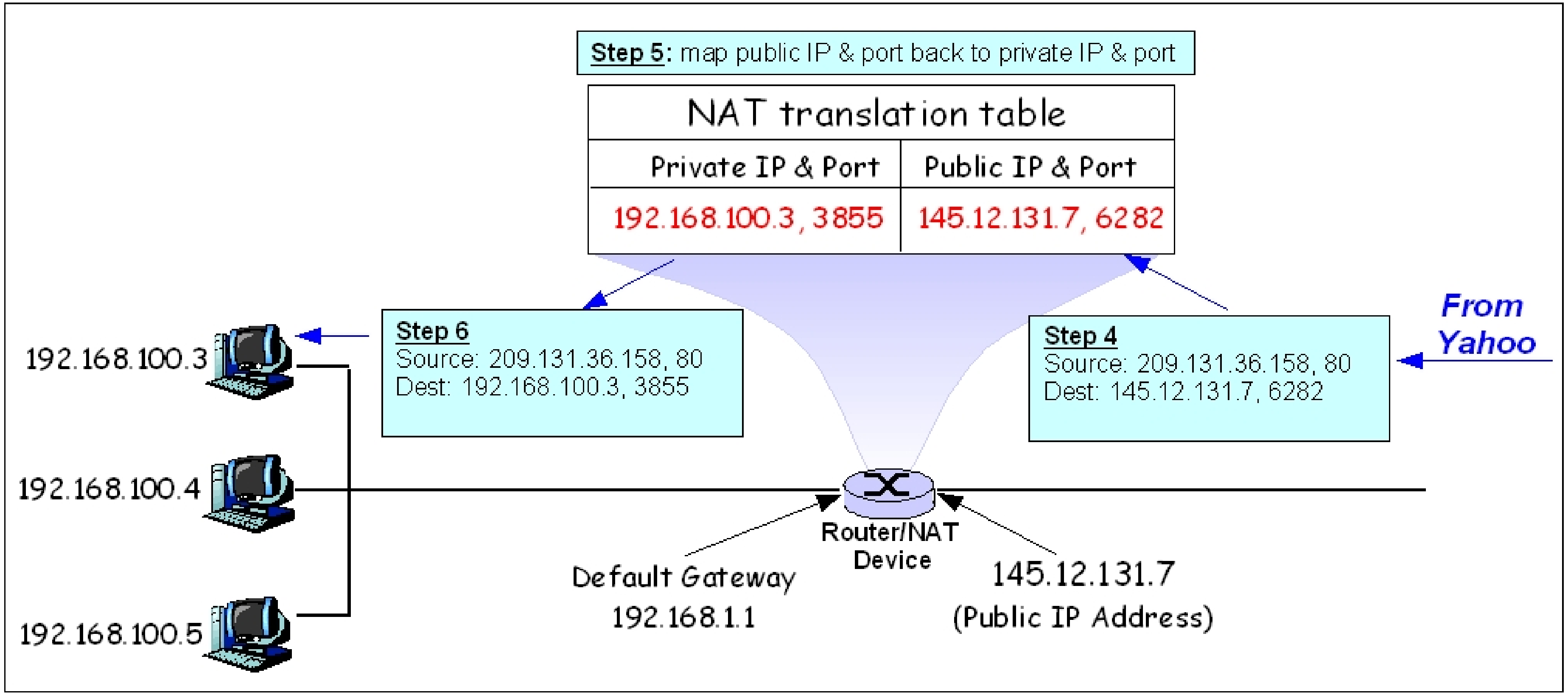
Translation



NAT Table with IP address & Port



NAT Table with IP address & Port



Multiple Public IPs can be used if needed to support more flows.

Concerns about NAT

- **Performance:**
  - Modifying the IP header by changing the IP address requires that NAT boxes recalculate the IP header checksum
  - Modifying port number requires that NAT boxes recalculate TCP checksum

- **End-to-end connectivity:**
  - NAT destroys universal end-to-end reachability of hosts on the Internet.
  - A host in the public Internet often cannot initiate communication to a host in a private network.
  - The problem is worse, when two hosts that are in a private network need to communicate with each other.

Address Resolution Protocol (ARP)

Address Mapping Cont..

- **The delivery of a packet to a host or a router requires two levels of addressing: *logical* and *physical*.**

- **We need to be able to map a logical address to its corresponding physical address and vice versa.**

- **Anytime a host or a router has an IP datagram to send to another host or router, it has the logical (IP) address of the receiver.**

- **But the IP datagram must be encapsulated in a frame to be able to pass through the physical network.**

- **This means that the sender needs the physical address of the receiver.**

- **A mapping corresponds a logical address to a physical address. ARP accepts a logical address from the IP protocol, maps the address to the corresponding physical address and pass it to the data link layer.**

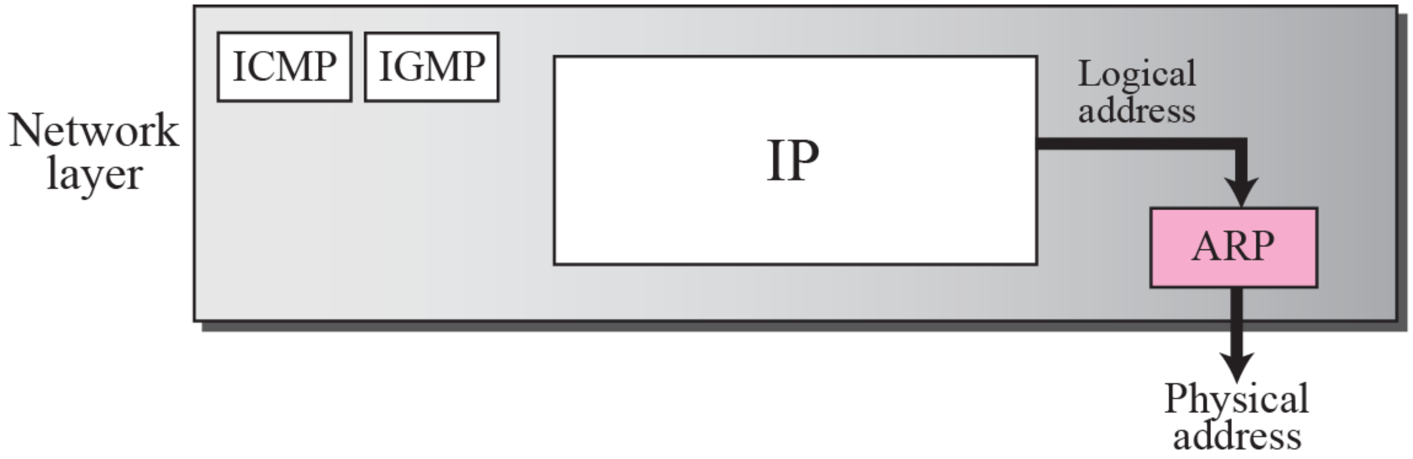
Logical and Physical addresses

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : iitbhilai.ac.in
   Description . . . . . . . . . . . : Qualcomm QCA9377 802.11ac Wireless Adapter
   Physical Address. . . . . . . . . : B0-68-E6-82-D9-8D
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::80d1:147e:1fc0:c043%9(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.3.54.107(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.192.0
   Lease Obtained. . . . . . . . . . : 29 September 2020 09:39:45
   Lease Expires . . . . . . . . . . : 29 September 2020 18:28:38
   Default Gateway . . . . . . . . . : 10.3.0.1
   DHCP Server . . . . . . . . . . . : 10.1.72.7
   DHCPv6 IAID . . . . . . . . . . . : 162556134
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-24-F3-49-C3-D8-D0-90-5B-50-B5
   DNS Servers . . . . . . . . . . . : 192.168.10.87
                                       192.168.10.72
```

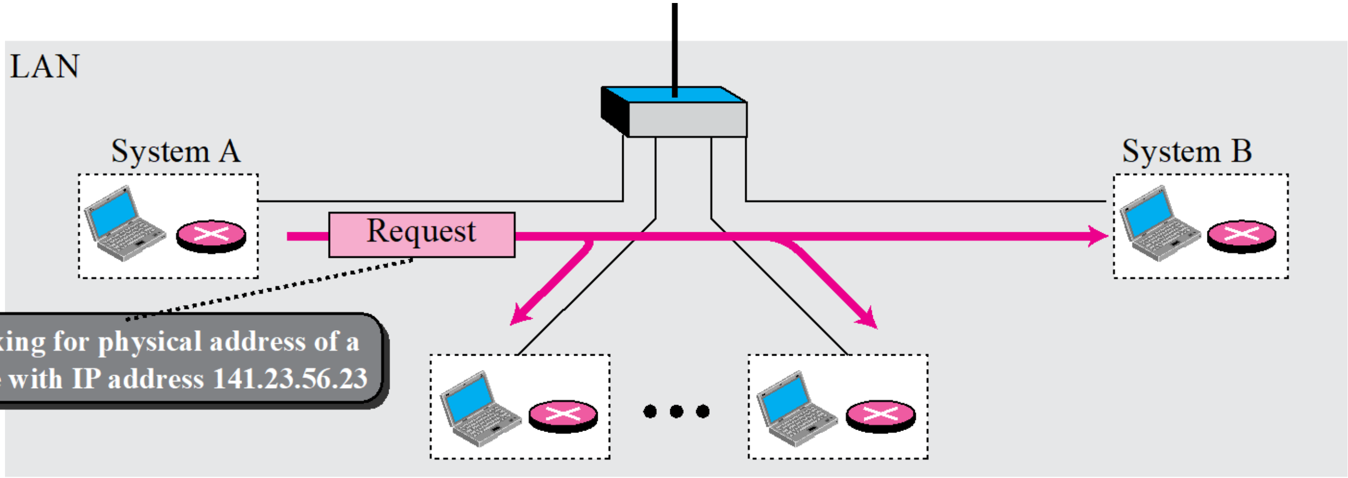
ARP Cache

```
C:\Users\Anand>arp -a

Interface: 10.3.54.107 --- 0x9
  Internet Address      Physical Address      Type
  10.3.0.1              00-00-0c-07-ac-cd     dynamic
  10.3.52.151           a4-fc-77-04-20-43     dynamic
  10.3.58.183           00-28-f8-93-47-7a     dynamic
  10.3.63.255           ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

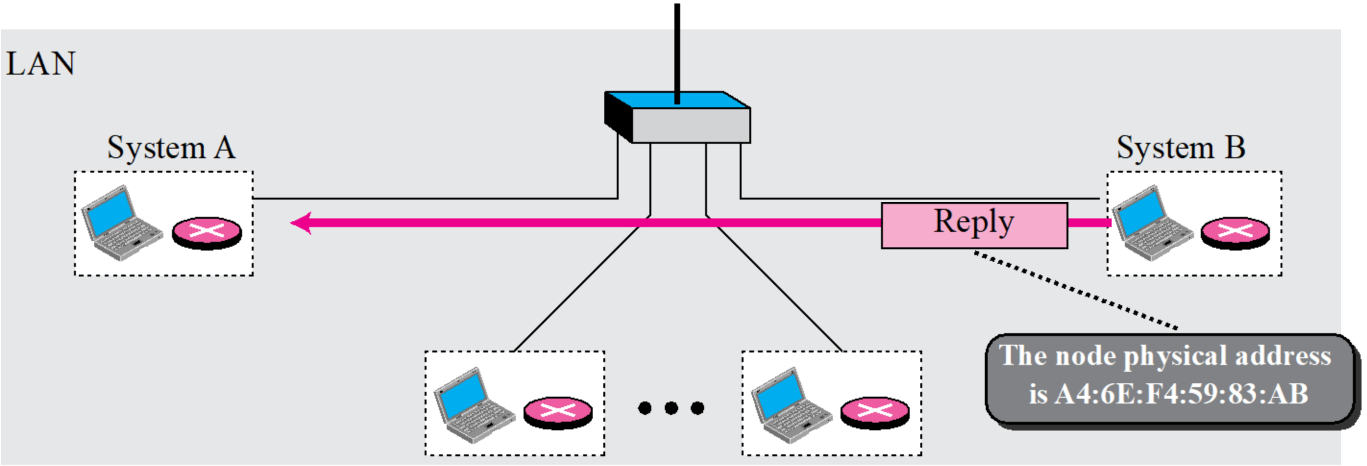
Position of ARP in TCP/IP protocol suite



ARP operation



a. ARP request is broadcast

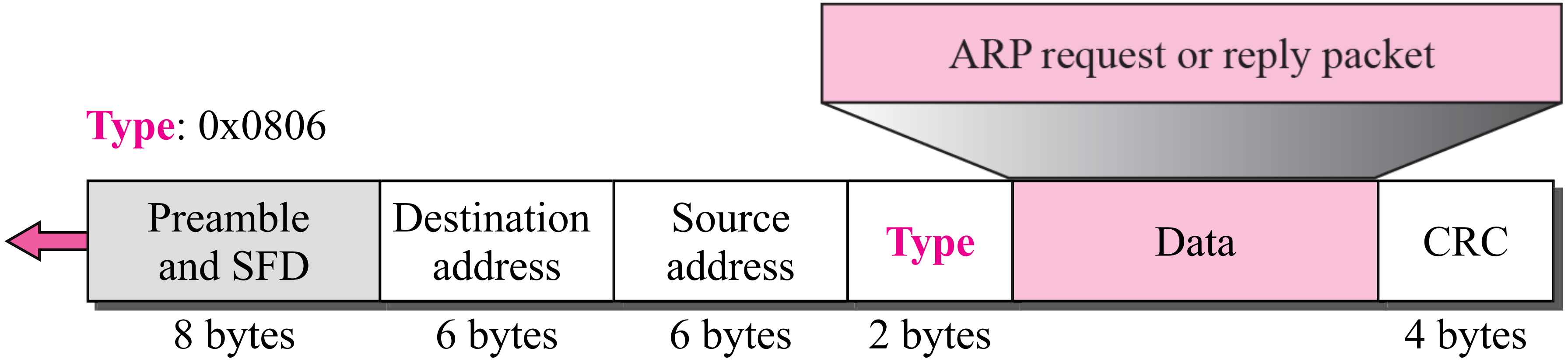


b. ARP reply is unicast

ARP packet

<table>
<tr><td colspan="2">Hardware Type</td><td>Protocol Type</td></tr>
<tr><td>Hardware length</td><td>Protocol length</td><td>Operation
Request 1, Reply 2</td></tr>
<tr><td colspan="3">Sender hardware address
(For example, 6 bytes for Ethernet)</td></tr>
<tr><td colspan="3">Sender protocol address
(For example, 4 bytes for IP)</td></tr>
<tr><td colspan="3">Target hardware address
(For example, 6 bytes for Ethernet)
(It is not filled in a request)</td></tr>
<tr><td colspan="3">Target protocol address
(For example, 4 bytes for IP)</td></tr>
</table>

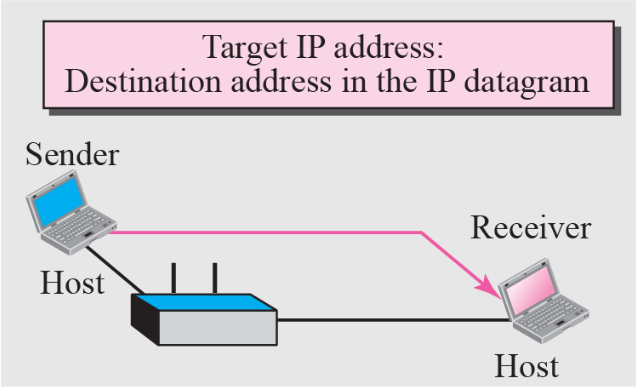
Encapsulation of ARP packet



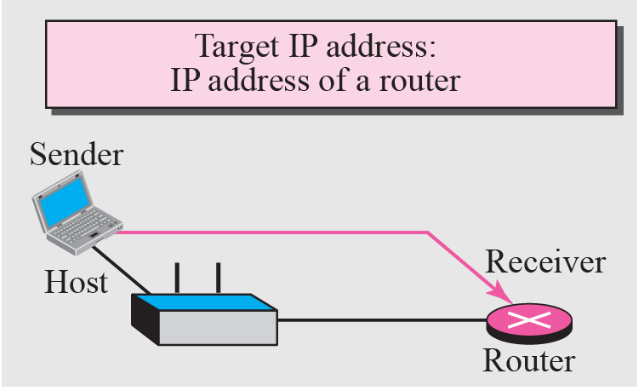
An ARP request is broadcast;
an ARP reply is unicast.

Four cases using ARP

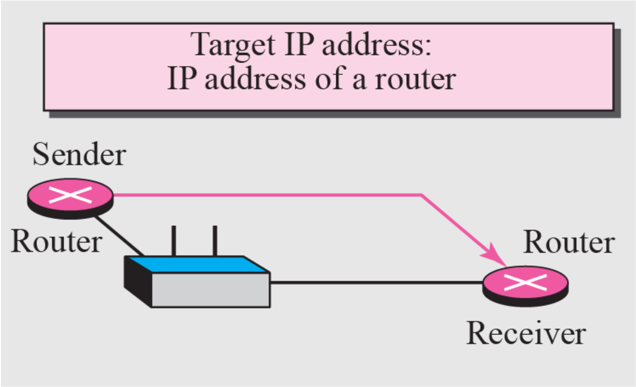
Case 1: A host has a packet to send to a host on the same network.



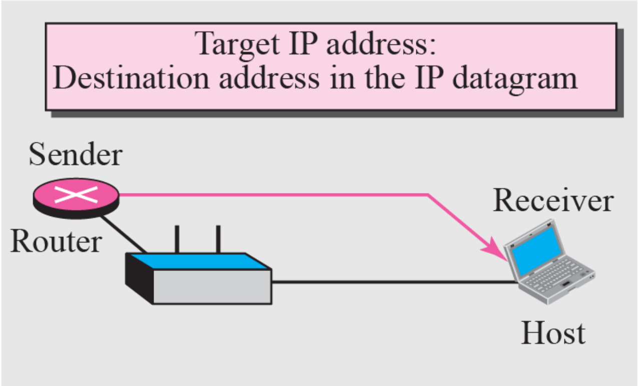
Case 2: A host has a packet to send to a host on another network.



Case 3: A router has a packet to send to a host on another network.



Case 4: A router has a packet to send to a host on the same network.



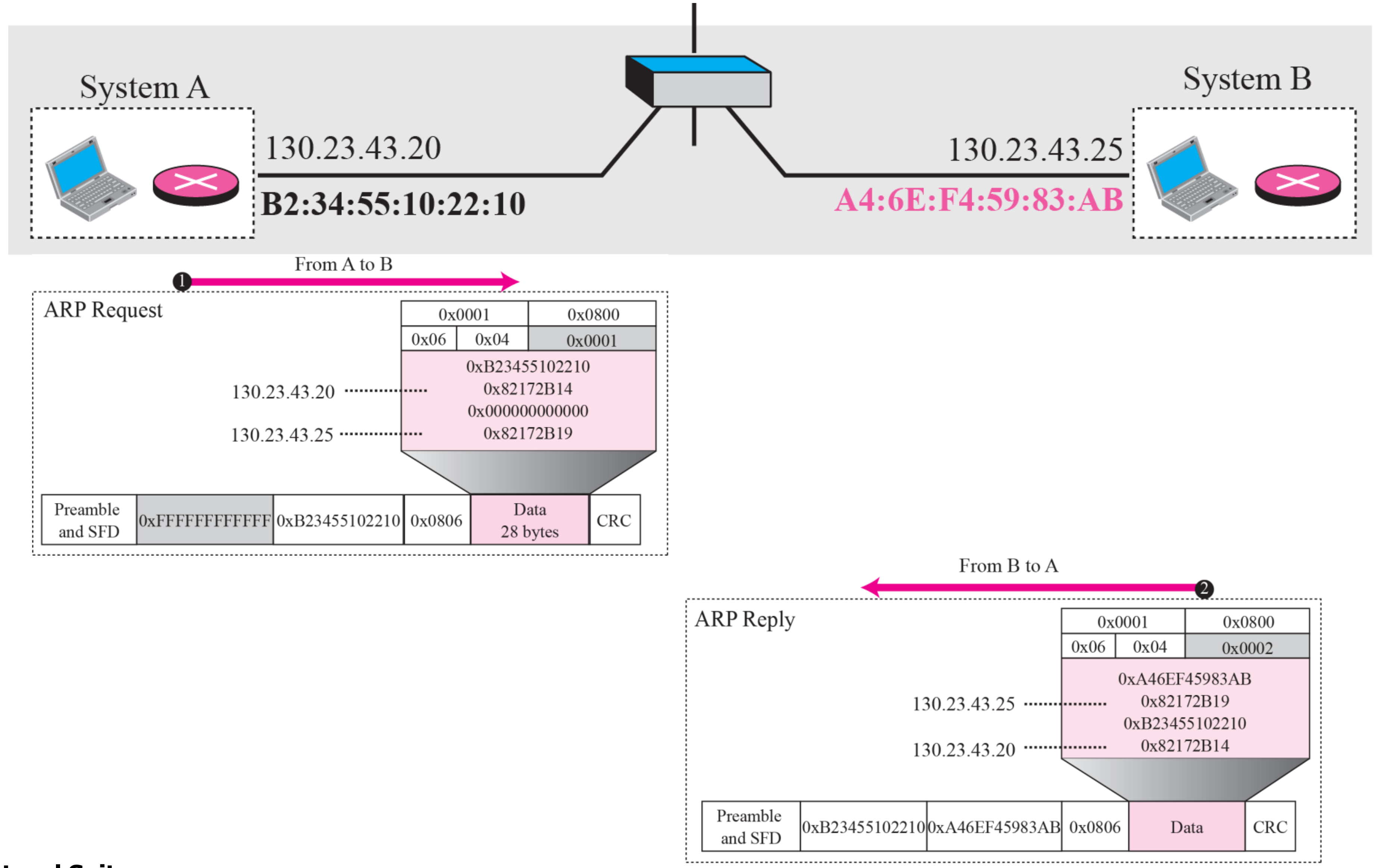
Example

A host with IP address 130.23.43.20 and physical address B2:34:55:10:22:10 has a packet to send to another host with IP address 130.23.43.25 and physical address A4:6E:F4:59:83:AB. The two hosts are on the same Ethernet network. Show the ARP request and reply packets encapsulated in Ethernet frames.

Solution

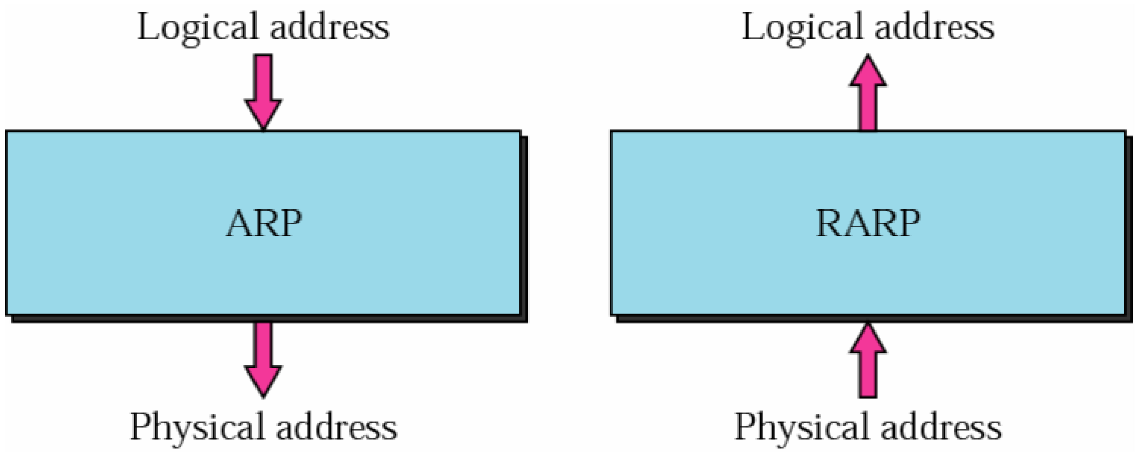
Figure shows the ARP request and reply packets. Note that the ARP data field in this case is 28 bytes, and that the individual addresses do not fit in the 4-byte boundary. That is why we do not show the regular 4-byte boundaries for these addresses. Also note that the IP addresses are shown in hexadecimal.

Example

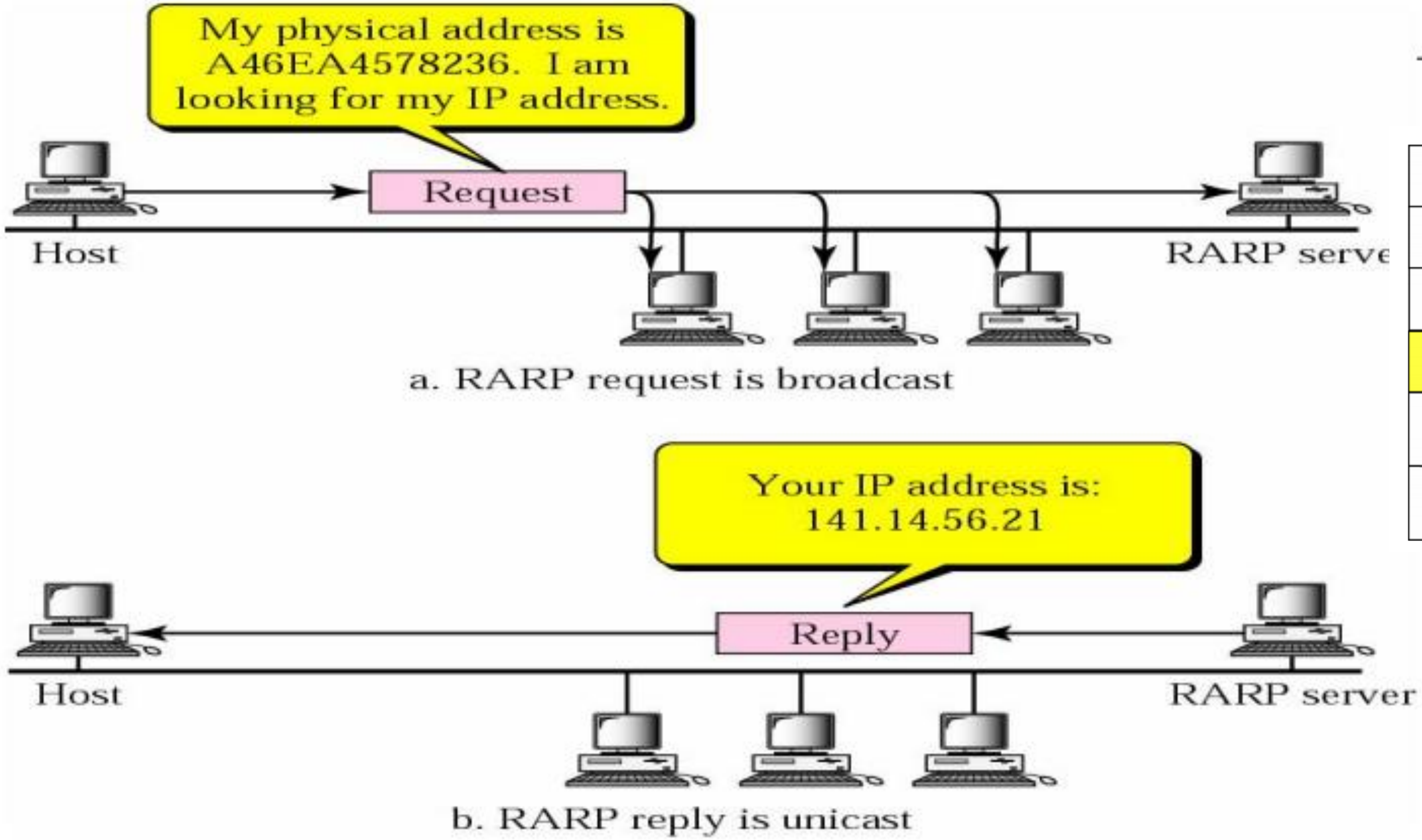

System A
130.23.43.20
B2:34:55:10:22:10
System B
130.23.43.25
A4:6E:F4:59:83:AB
From A to B
ARP Request
0x0001
0x0800
0x06
0x04
0x0001
0xB23455102210
0x82172B14
0x000000000000
0x82172B19
130.23.43.20
130.23.43.25
Preamble and SFD
0xFFFFFFFFFFFF
0xB23455102210
0x0806
Data 28 bytes
CRC
From B to A
ARP Reply
0x0001
0x0800
0x06
0x04
0x0002
0xA46EF45983AB
0x82172B19
0xB23455102210
0x82172B14
130.23.43.25
130.23.43.20
Preamble and SFD
0xB23455102210
0xA46EF45983AB
0x0806
Data
CRC

RARP

- Physical address to logical address.
- A diskless machine is usually booted from ROM. RARP is used for diskless machine which can not store the IP address.
- Request Broadcast, reply unicast by RARP server.



RARP Operation



Hardware type		Protocol type
Hardware length	Protocol length	Operation Request 3, Reply 4
Sender hardware address (For example, 6 bytes for Ethernet)		
Sender protocol address (For example, 4 bytes for IP) (It is not filled for request)		
Target hardware address (For example, 6 bytes for Ethernet) (It is not filled for request)		
Target protocol address (For example, 4 bytes for IP) (It is not filled for request)		

The format of the RARP packet is the same as the ARP packet,
Except that the operation field is three for RARP request
message and Four for RARP reply message

Alternative Solution to RARP

- When a diskless computer is booted, it needs more information in addition to its IP address
  - like subnet mask,
  - default gateway/router,
  - DNS server,

- RARP cannot provide this extra information

- Hence we need something more than RARP i.e., DHCP.